

Body-fitted DFG cylinder verification

Abstract

Three actual polygon-grid simulations assess steady laminar external flow. Numerical convergence and strict physical error below 3% are separate gates. Complete raw fields, force components, histories, independent NumPy equation audits and source hashes are supplied. Failed metrics remain failed.

1. Physical problem

DFG 2D-1: $Re=20$, $\rho=1 \text{ kg/m}^3$, $\mu=0.001 \text{ Pa s}$, channel $2.2 \times 0.41 \text{ m}$, cylinder center $(0.2, 0.2) \text{ m}$ and diameter 0.1 m . Parabolic inlet has mean speed 0.2 m/s . Walls are no-slip; outlet pressure is zero.

2. Method and mesh

Collocated SIMPLE uses first-order upwind momentum transport, nonorthogonal diffusion and Rhie-Chow face flux. Walls are polygon chords. CPU SuperLU solves the same finite-volume linear equations with true residual checks. A full-grid coupled Picard residual correction retains periodic connectivity and exact assembled operator probes. Final residuals are recomputed after correction; physical flux defect must be below $1e-7$.

3. Acceptance and reproducibility

Each integrated physical reference error must be strictly below 3%; steady momentum, continuity and boundary imbalance must be below $1e-6$. Independent NumPy gradients, complete momentum residuals, mass and inertia-free physical Rhie-Chow flux are audited. Grid refinement alone does not prove an asymptotic regime. Exact commands and editable LaTeX source accompany this PDF. Elapsed solve time excludes mesh setup, export and plotting; no matched TensorLBM performance comparison is claimed.

4. Reference integrity

FeatFlow DFG 2D-1 high-accuracy tabulated values: $C_d=5.57953523384$, $C_l=0.010618948146$, front/rear pressure difference= 0.11752016697 Pa .

5. Results and independent audits

CPU float64; three actual grids. Physical errors are evaluated separately.

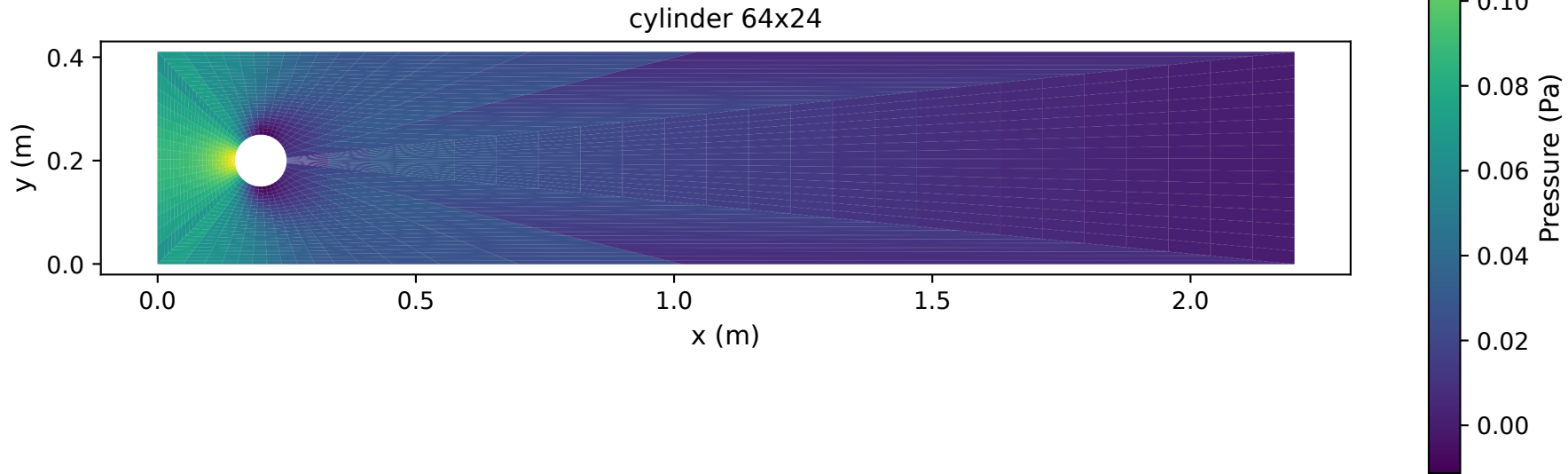
Grid	Cells	Steps	Steady	Cd	Cl	Max physical error (%)	Qualified
64x24	1536	28	True	6.324791	0.01510167	42.214	False
128x48	6144	35	True	5.972001	0.009706585	8.592	False
256x96	24576	33	True	5.804957	0.007083128	33.297	False

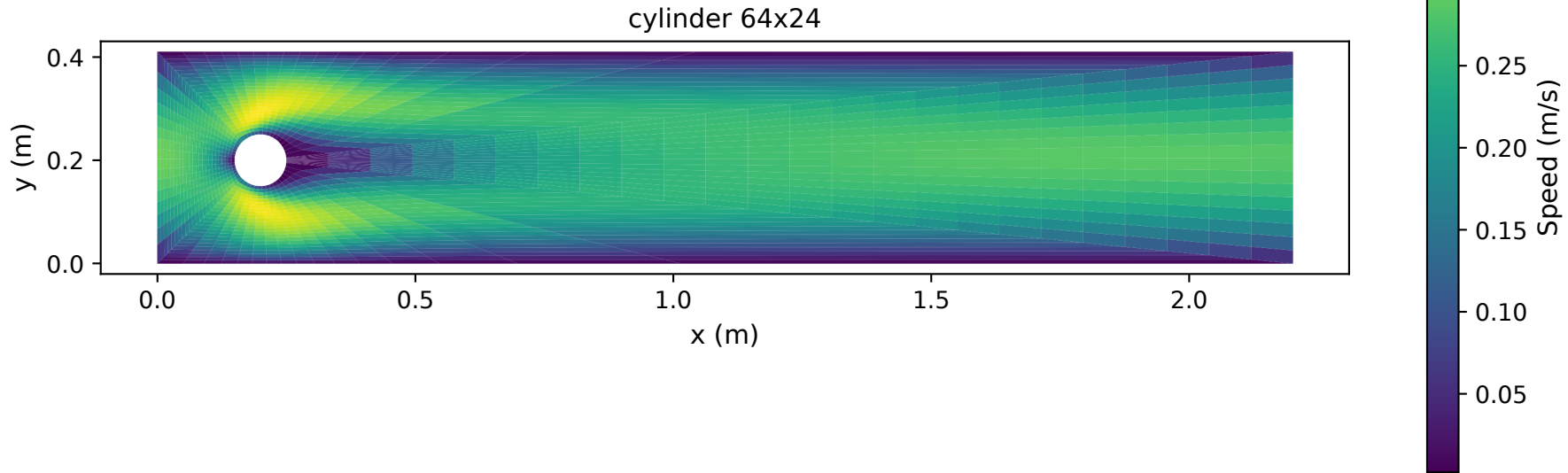
Independent equations reconstructed from saved fields

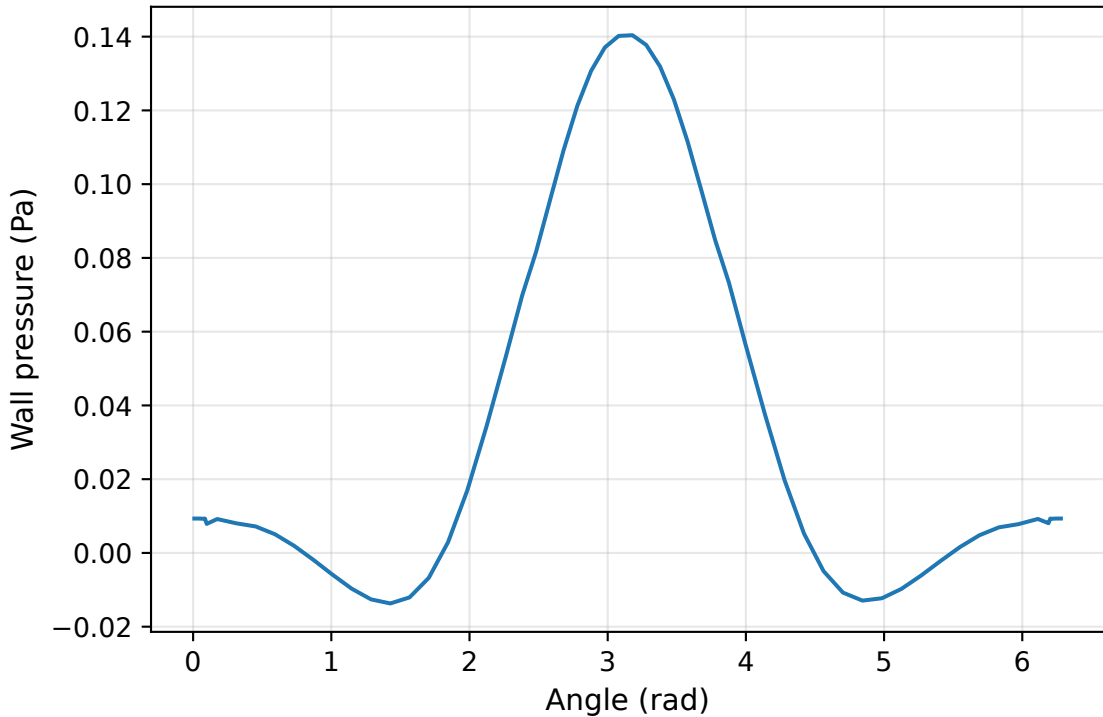
Grid	Momentum	Continuity	Physical flux defect	Fixed point
64x24	9.382e-07	2.253e-09	3.255e-10	True
128x48	9.921e-07	8.649e-10	1.765e-10	True
256x96	9.873e-07	4.001e-11	7.821e-12	True

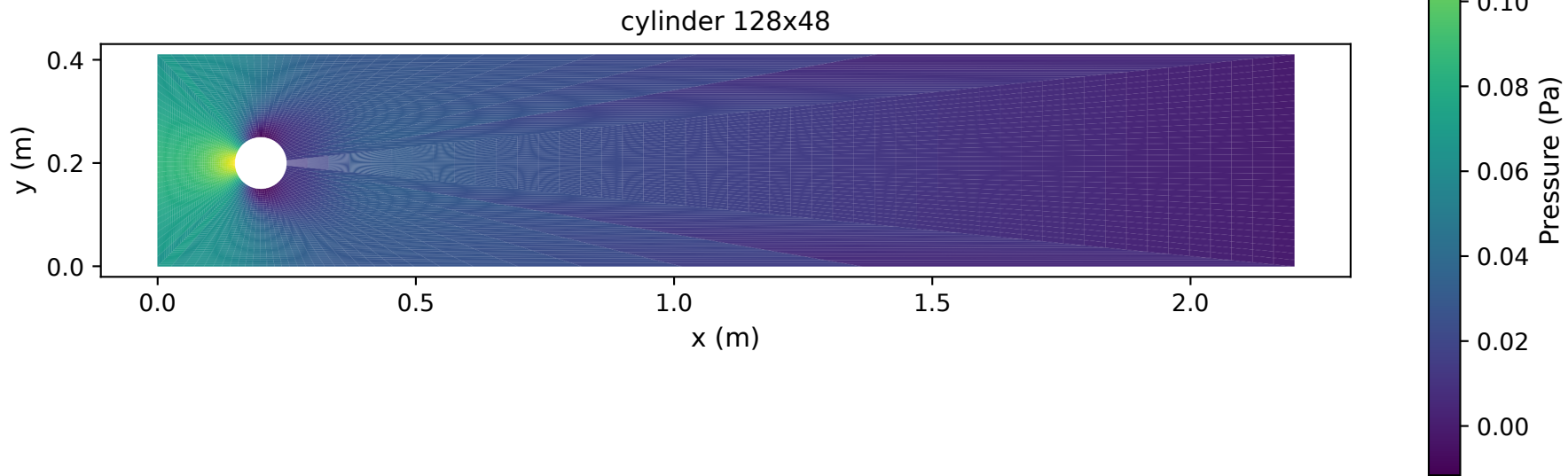
6. Discussion and limits

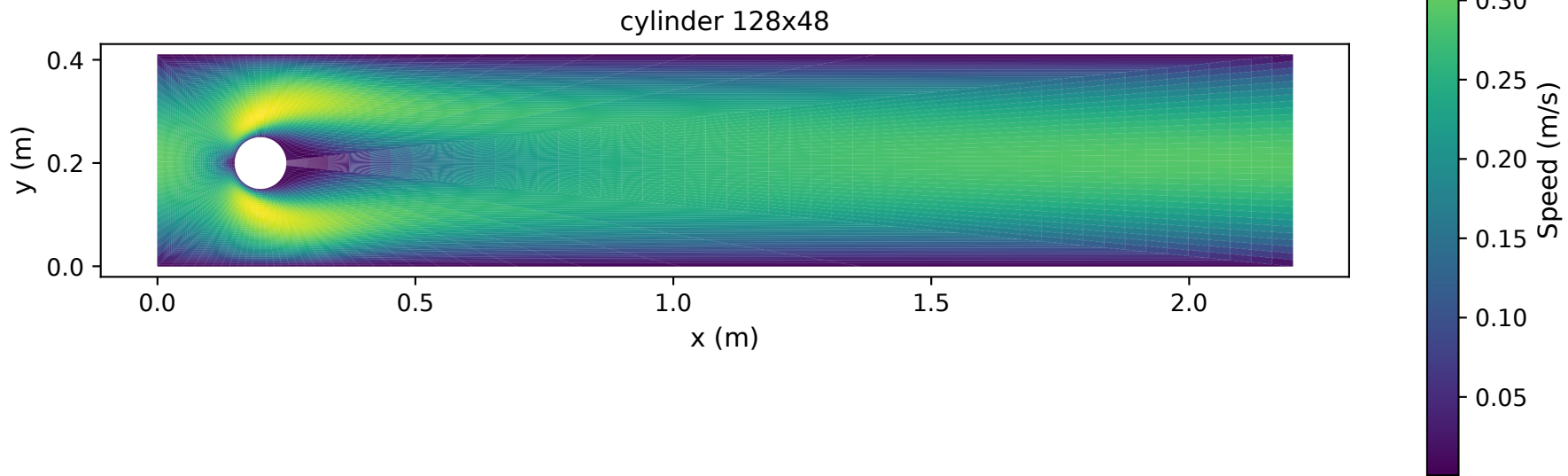
All physical gates must pass independently. Iterative convergence cannot replace physical accuracy. The finest mesh has 24576 control volumes and 73728 primary velocity/pressure unknowns. First-order transport, wall traction, polygon geometry and finite-domain effects require further separation before qualification.



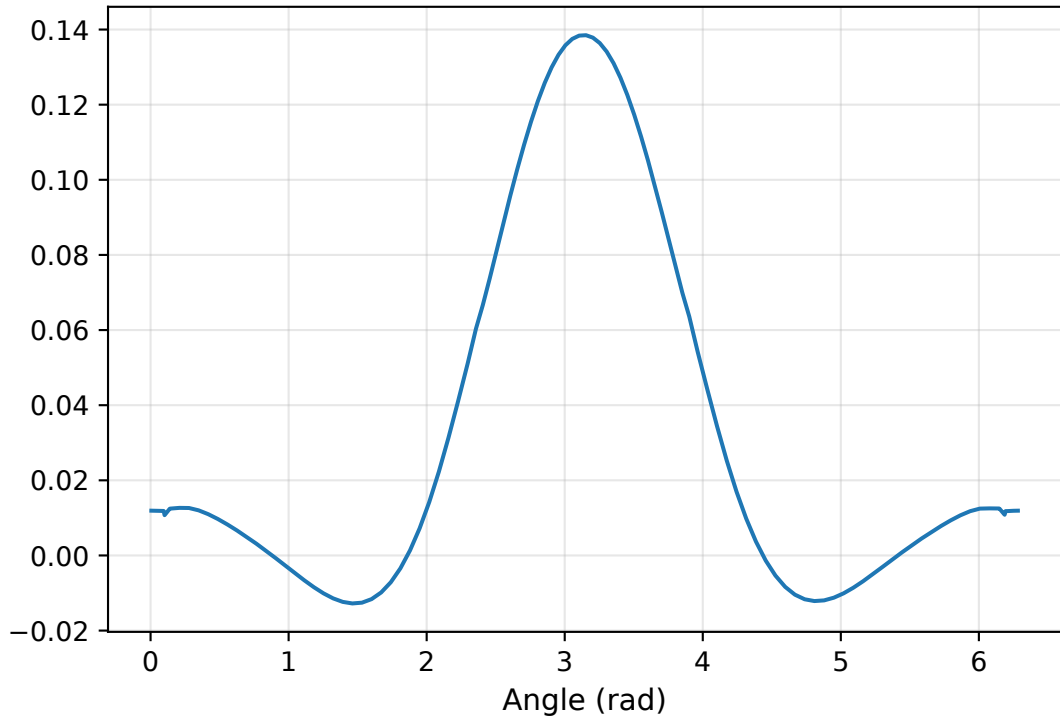


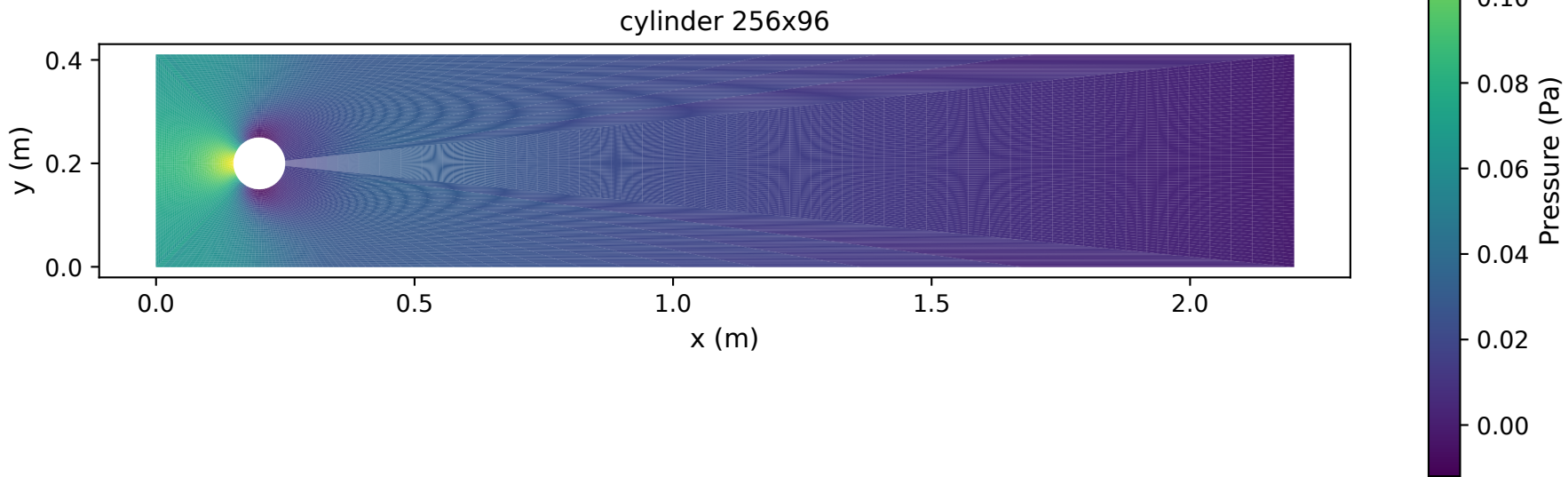


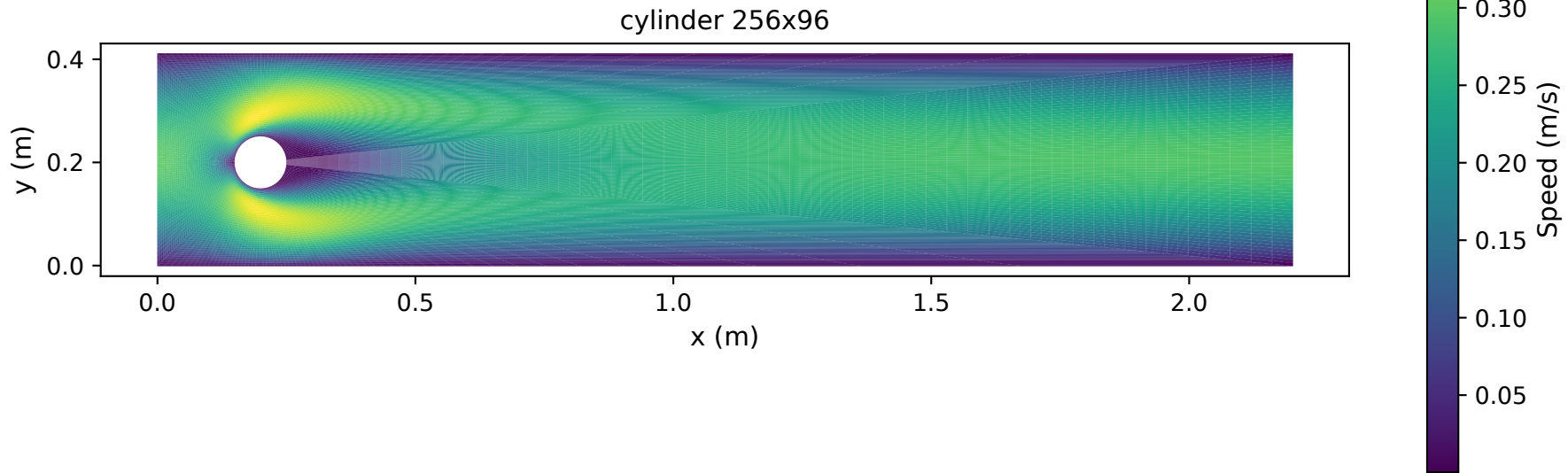




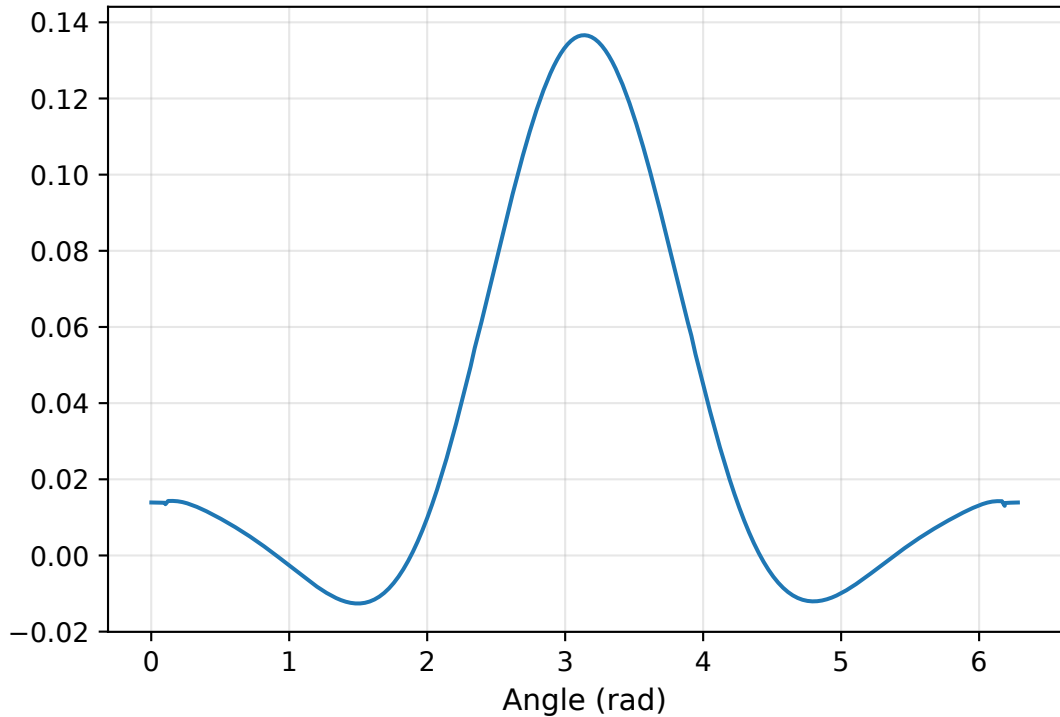
Wall pressure (Pa)



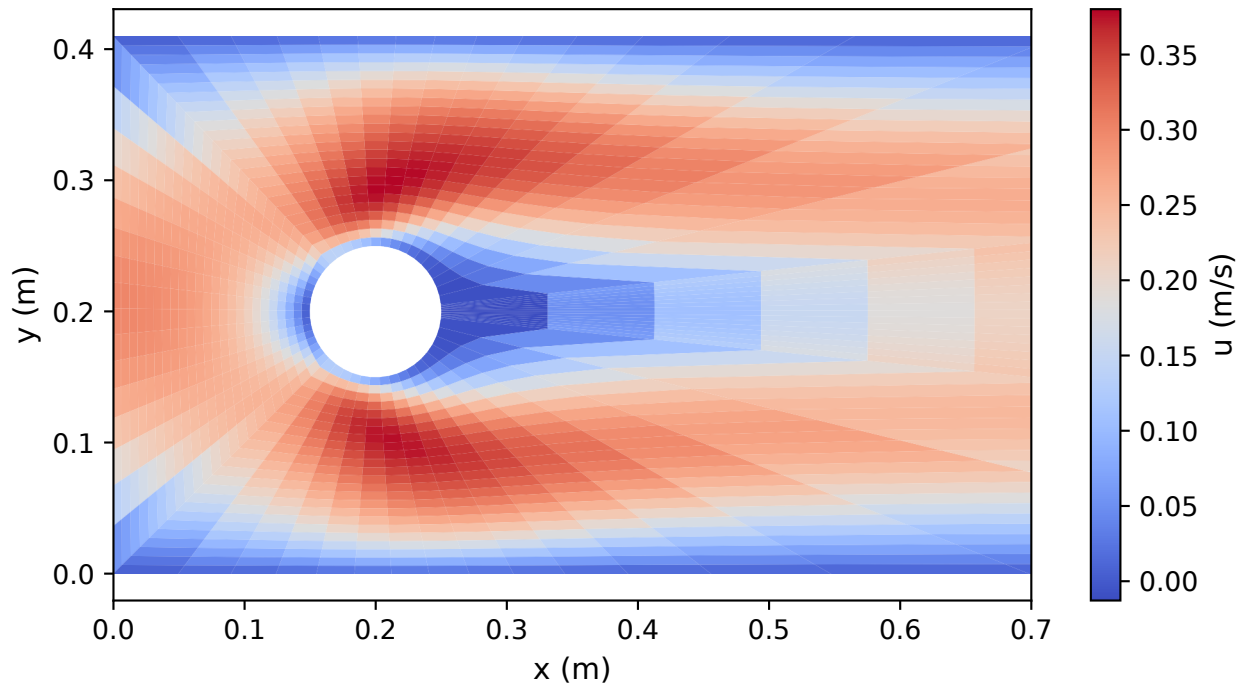




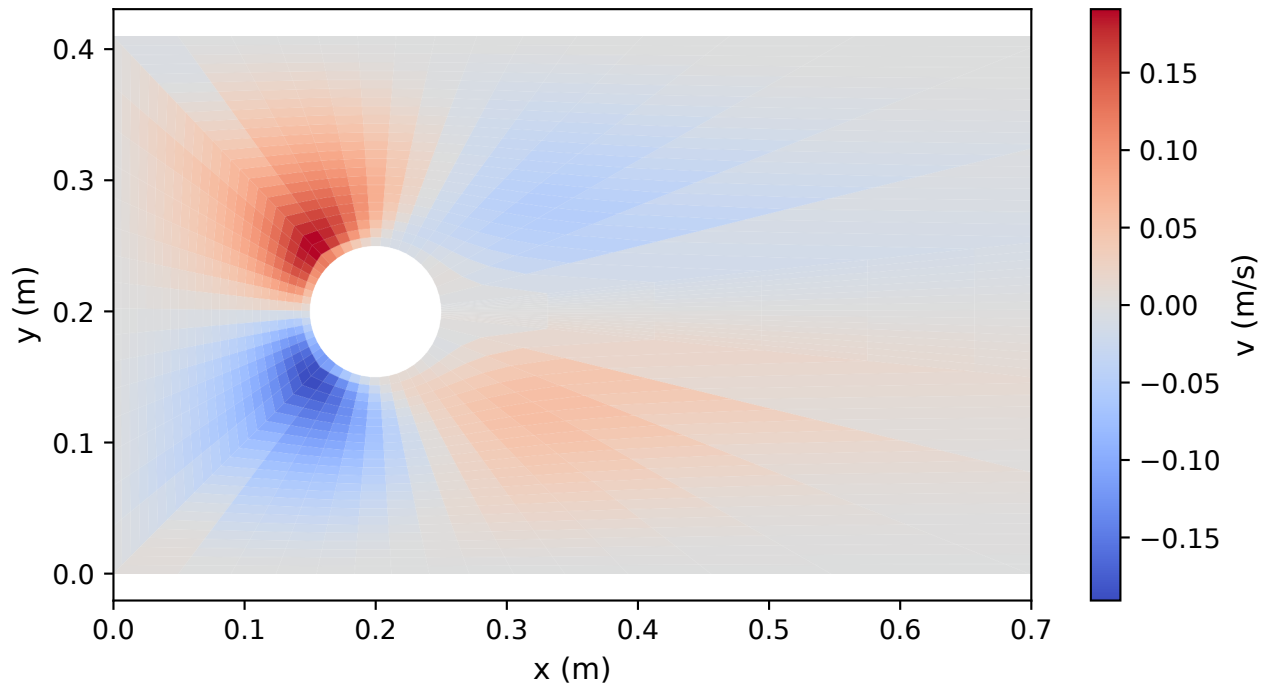
Wall pressure (Pa)



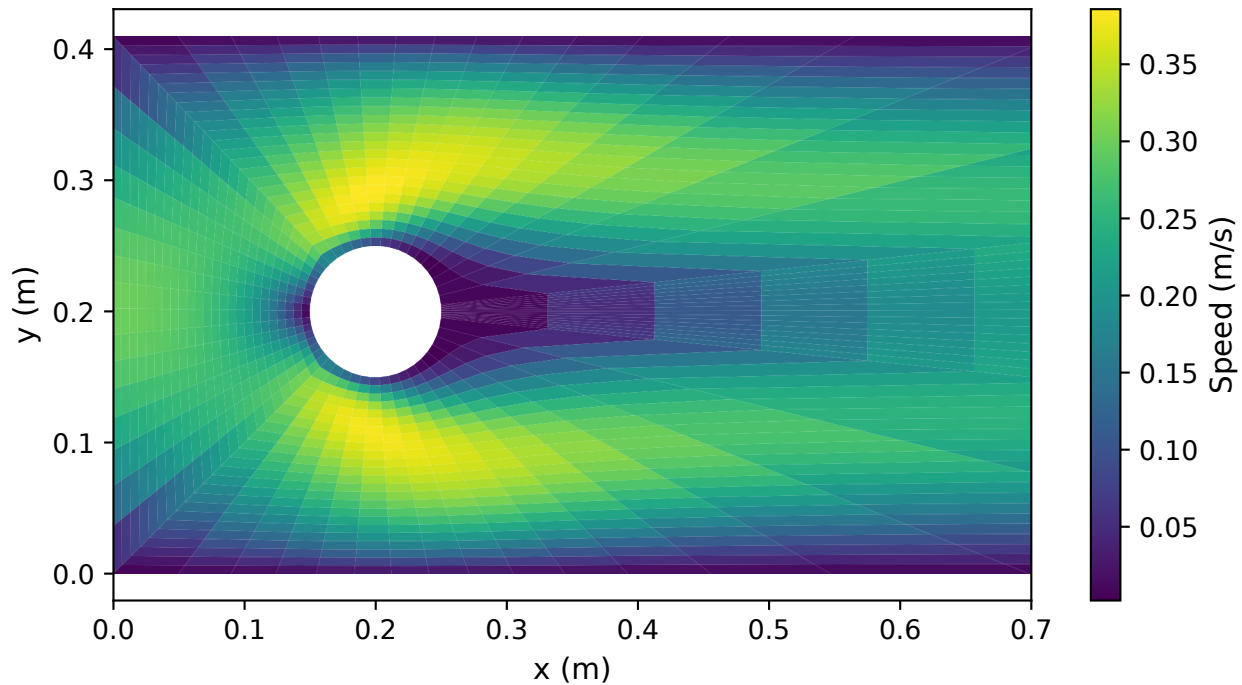
cylinder 64x24



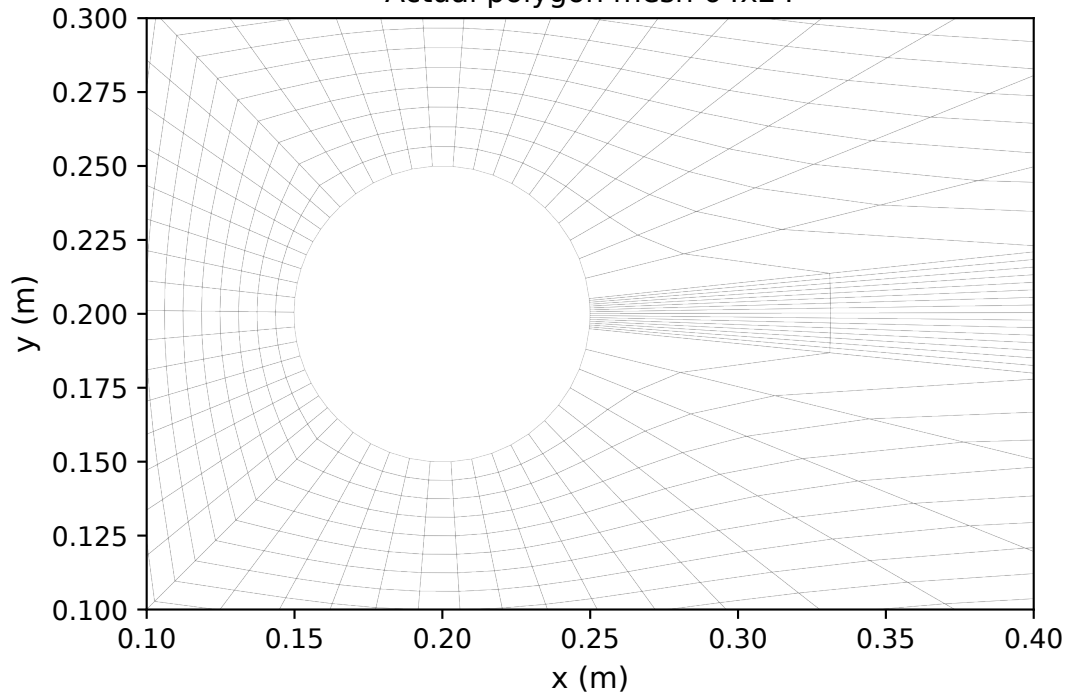
cylinder 64x24



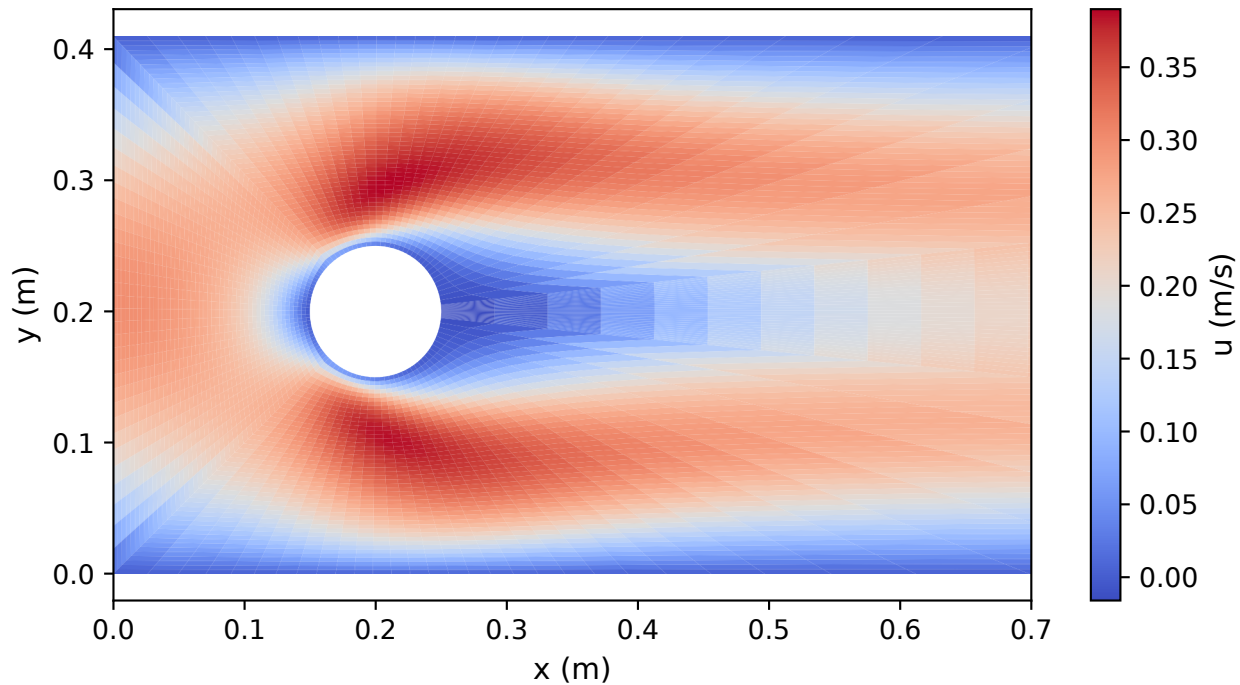
Wall and wake detail: 64x24



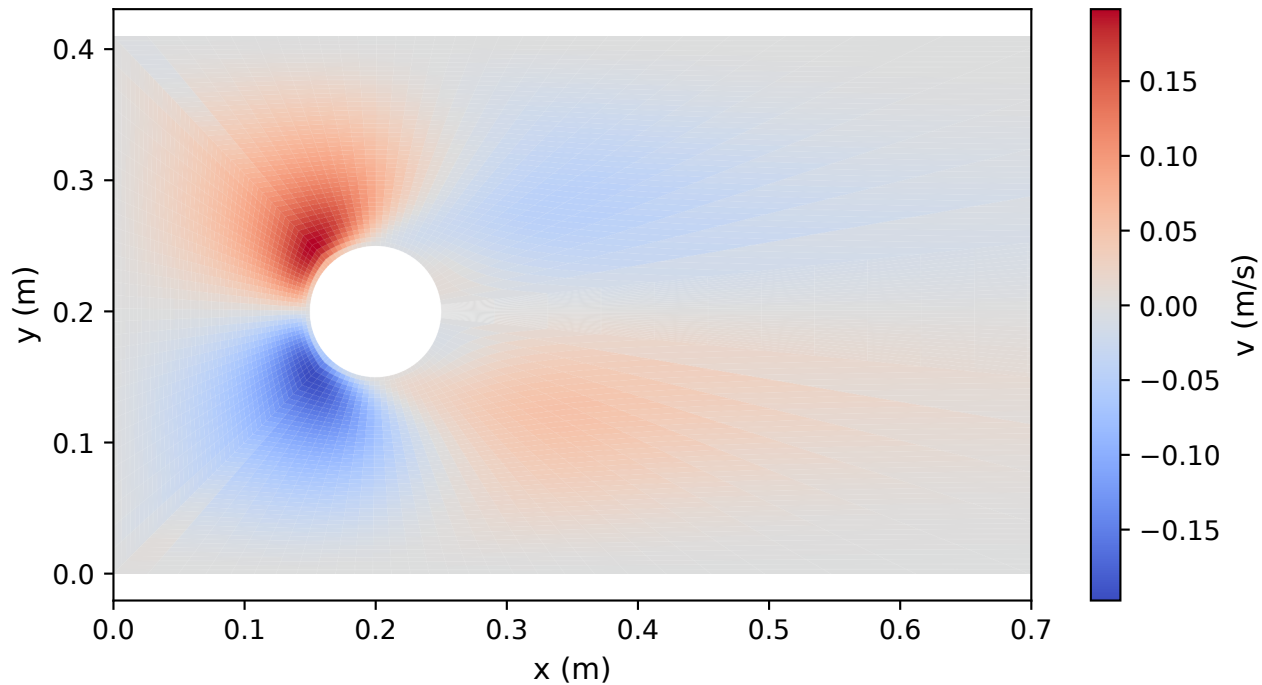
Actual polygon mesh 64x24



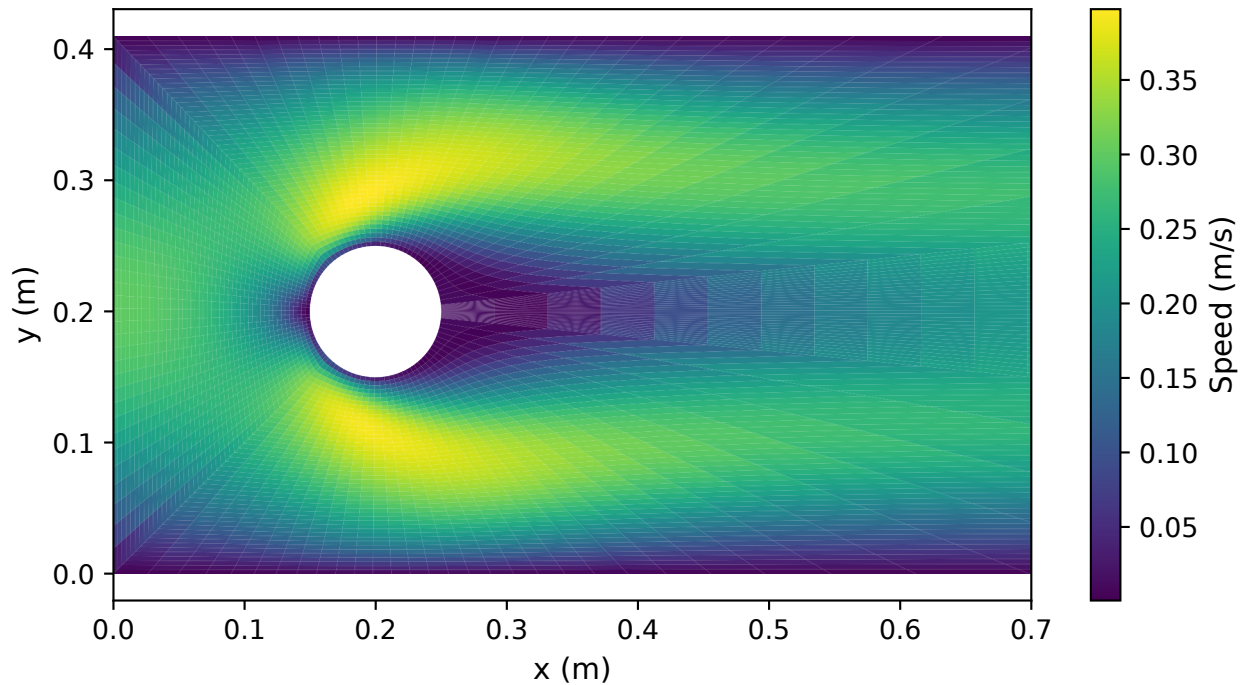
cylinder 128x48



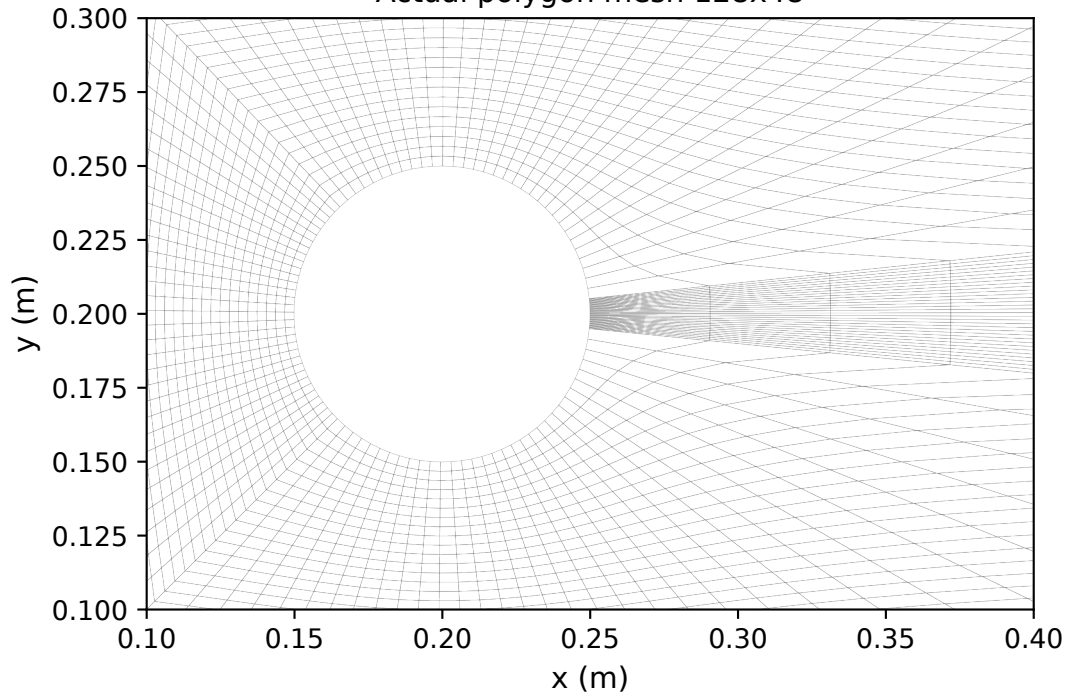
cylinder 128x48



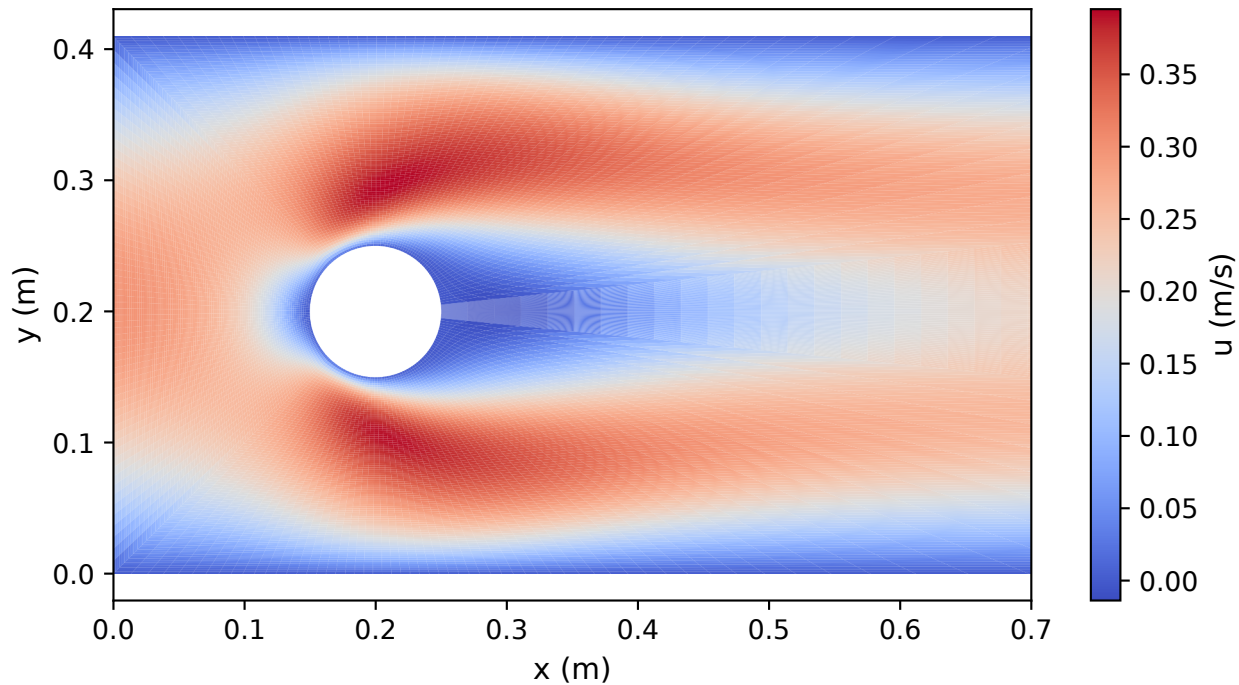
Wall and wake detail: 128x48



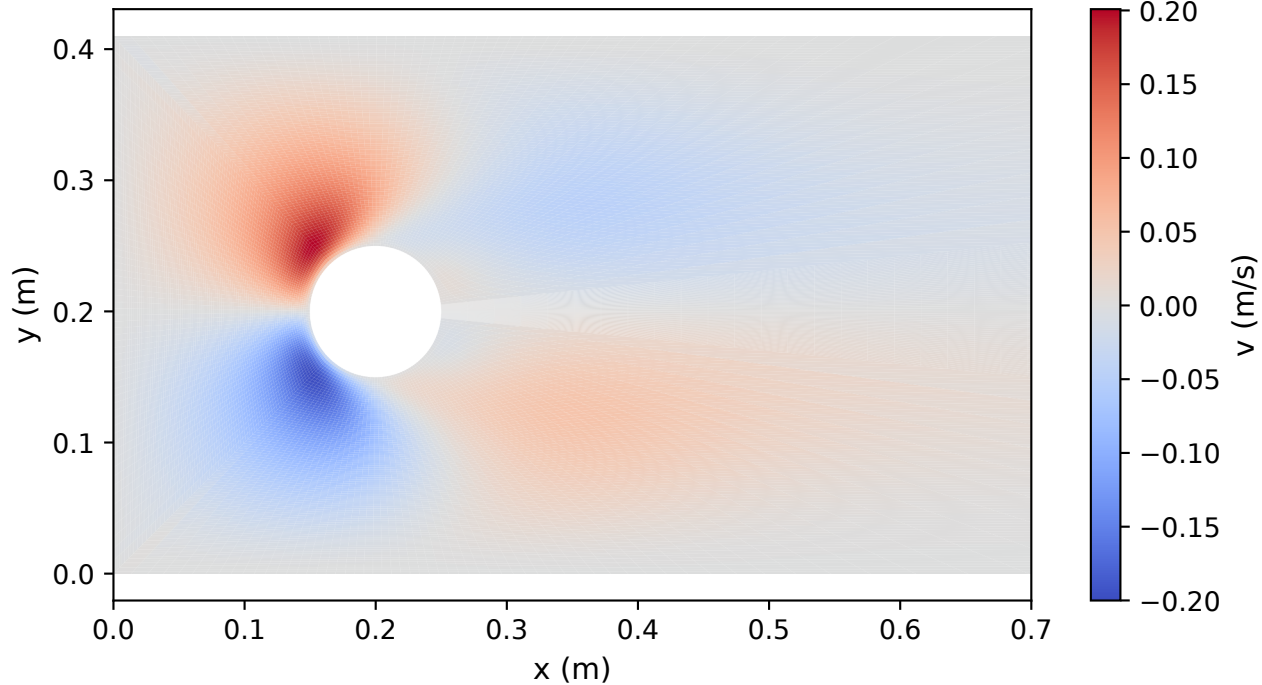
Actual polygon mesh 128x48



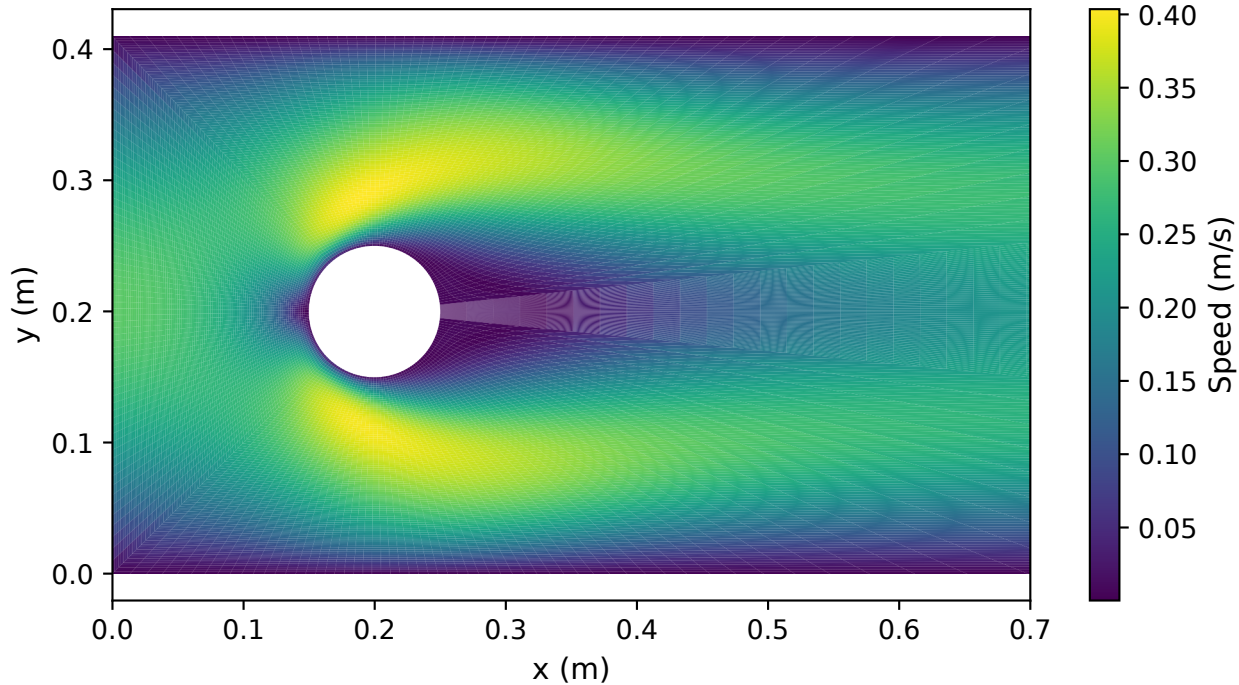
cylinder 256x96



cylinder 256x96



Wall and wake detail: 256x96



Actual polygon mesh 256x96

